Wei-Jhih Huang

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Research Interests

AI-Native Radio Access Networks, Learning-Based Resource Orchestration, Deep Learning for Cross-Domain Applications

Education

Bachelor, Communication Engineering, National Central University — ***Aug 2022 – Present***
GPA: 3.94/4.3, Expected Graduation: July 2026 — Taoyuan, Taiwan

Research Experience

Wireless Mobile Network Lab, National Taiwan University — ***Apr 2026 – Present***
Research Assistant, Advisor: **Prof. Hung-Yu Wei** — Taipei, Taiwan

- Developed an AI-native orchestration framework for LLM deployment across distributed edge networks, resolving end-to-end latency by strategically balancing communication overhead with computational resource availability.
- Designed predictive resource management algorithms within a 6G AI-RAN architecture, utilizing cross-layer telemetry to anticipate network dynamics and enable proactive data staging for seamless service continuity in highly mobile scenarios.

Information Processing and Communications Lab, National Central University — ***Feb 2024 – Present***
Undergraduate Research Student, Advisor: **Prof. Chih-Wei Huang** — Taoyuan, Taiwan

- Proposed a self-aware Meta-RL framework for 6G radio resource management with a Feature-wise Attention Gate enabling KNN-based OOD detection of traffic shifts; achieved ***0.790*** AUROC and ***1.51 ms*** latency on 100 3GPP 5QI-compliant scenarios.
- Built a trajectory-embedding mechanism for scenario distribution-aware DRL fine-tuning, reducing unnecessary DRL retraining by ***90%*** and improving user quality of service by ***11%*** in cross-layer resource management scenarios.

Electronic Design Automation Lab, The Chinese University of Hong Kong — ***Jul – Aug 2025***
Summer Visiting Research Student, Advisor: **Prof. Bei Yu** — Hong Kong

- Enhanced *NUA-Timer++* for graph-based hierarchical circuit timing prediction under Non-Uniform Input Arrival Times, achieving a ***1.96x*** improvement in consistency and reducing average error to ***0.42x*** compared to SOTA.
- Collaborated with a postdoctoral researcher on Transformer architecture design and loss function refinement for *NUA-Timer++*; independently investigated a critical-path-based timing constraint extension as an auxiliary signal for logic synthesis.

Publications

- **Wei-Jhih Huang**, Jin-Min Yang, Ning Ma, Sheng-Liang Wu, Chih-Wei Huang. "Deviation-Aware Trajectory Embedding for Fine-Tuning Triggering in DRL-Based Wireless Network Resource Management," *In Proceedings of the IEEE Wireless Communications and Networking Conference (WCNC), 2026.*
- Ziyi Wang, **Wei-Jhih Huang**, Fangzhou Liu, Tsung-Yi Ho, David Z. Pan, Bei Yu. "NUA-Timer++: Timing Prediction for Hierarchical RTL under Non-Uniform Input Arrival Times," *IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems (TCAD)*, Under Review.
- Ching-Wei Lin, **Wei-Jhih Huang**, Sheng-Liang Wu, Chih-Wei Huang, Phone Lin, Shun-Ren Yang. "PACI-Net: Physics-Aware Distillation for Explainable Radio Resource Management," *IEEE Global Communications Conference (GLOBECOM)*, Under Review.

Honors & Awards

- **Outstanding Student Scholarship,** National Central University, 2026
- **Outstanding College Youth Award,** School Level, Taiwan, 2026
- **Undergraduate Research Fellowship,** National Science and Technology Council (NSTC), Taiwan, 2025–2026
- **Honorable Mention,** Undergraduate Project Competition, Department of Communication Engineering, NCU, 2025
- **Merit Award,** Individual Service-Learning Excellence Scholarship, National Central University, 2025
- **High Distinction** and **Public Service Scholarship,** National Central University (as President, NCU Rural Service Club), 2024
- **Excellence Award,** NCU Club Evaluation (as Presenter and President, NCUgF Rural Service Club), 2024

Teaching & Work Experience

Chunghwa Telecom ***Sep – Dec 2025***
AI Software Development Intern Taipei, Taiwan

- Spearheaded the full-stack development of an intelligent project management system, spanning UI/UX design and n8n workflow backend orchestration, driven by RAG-based agentic models to enhance operational efficiency by ***33%.***

National Central University ***Feb – Jun 2025***
Teaching Assistant, Courses: ***Natural Language Processing*** *and* ***Enterprise Application of Pattern Recognition***

- Assisted in lecture preparation and grading for NLP topics for ***60*** students and image recognition topics for ***60*** students, analyzing the underlying AI technologies of real-world company products to support students in final project research.

Industrial Technology Research Institute ***Jun – Sep 2024***
AI Software Summer Intern Hsinchu, Taiwan

- Built and deployed an LLM-based service chatbot at *Taipei Songshan Sports Center*, integrating Azure OpenAI with Linux backend services to provide ***24/7*** automated customer support, reducing customer queue time by ***50%*** and improving service efficiency.